Amirkabir University of Technology
(Tehran Polytechnic)

Department of Computer Engineering

MINI-TASK 9
Background Sound Classification and Voice Activity Detection

VOiCES-subset ● SVM ● Energy-Based VAD ● Silero VAD

Course: Digital Speech Processing

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Abstract

I used VOiCES-subset to study recording-level background classification and voice activity detection (VAD). After excluding one non-audio archive entry, I processed 908 mono WAV recordings sampled at 16 kHz. I extracted 30 acoustic statistics and trained an RBF SVM. I obtained 69.42% test accuracy with whole-recording features and 87.60% when I used the 25% lowest-energy frames. I also improved the macro F1-score from 0.6901 to 0.8730. In five-fold speaker-grouped cross-validation on the training partition, I measured mean accuracies of 70.71% and 87.83%, respectively. For VAD, I compared an energy-based detector with pretrained Silero VAD on four examples and an extended sample of 40 recordings. I measured 81.70% mean frame agreement across the extended sample. I did not report this agreement as detection accuracy because I had no independently annotated speech intervals.

1. Task and scope

I addressed the two assigned topics using VOiCES-subset. For the event/background task, I classified each recording by its background condition: none, babble, music, or television. I did not attempt to identify the temporal boundaries of individual background sound events. For the VAD task, I identified probable speech intervals using two approaches and compared their predictions without independent reference annotations.

2. Dataset and preparation

I cloned the Lab41 VOiCES-subset repository into Google Colab and inspected eight TAR archives and three CSV metadata files. The archives contained 909 entries with .wav suffixes; one was an AppleDouble-style ._ entry that could not be opened as audio. I excluded that entry, retaining 908 valid mono recordings at 16,000 samples per second.

Background	Valid WAV recordings
No distractor	227
Babble	228
Music	231
Television	222
Total	908

I identified 104 speakers in the recordings. Their durations ranged from 2.66 to 19.10 s, with a mean of approximately 15.56 s. I used a speaker-grouped split with 666 recordings from 78 speakers for training and 242 recordings from 26 different speakers for testing. I checked the speaker IDs and found no overlap between the two sets.

3. Audio inspection

I loaded one representative WAV recording from each background condition and displayed its waveform and magnitude spectrogram. I used these examples to inspect acoustic differences, but I did not control their utterances, microphones, or rooms. I therefore did not attribute the visible differences exclusively to background class.

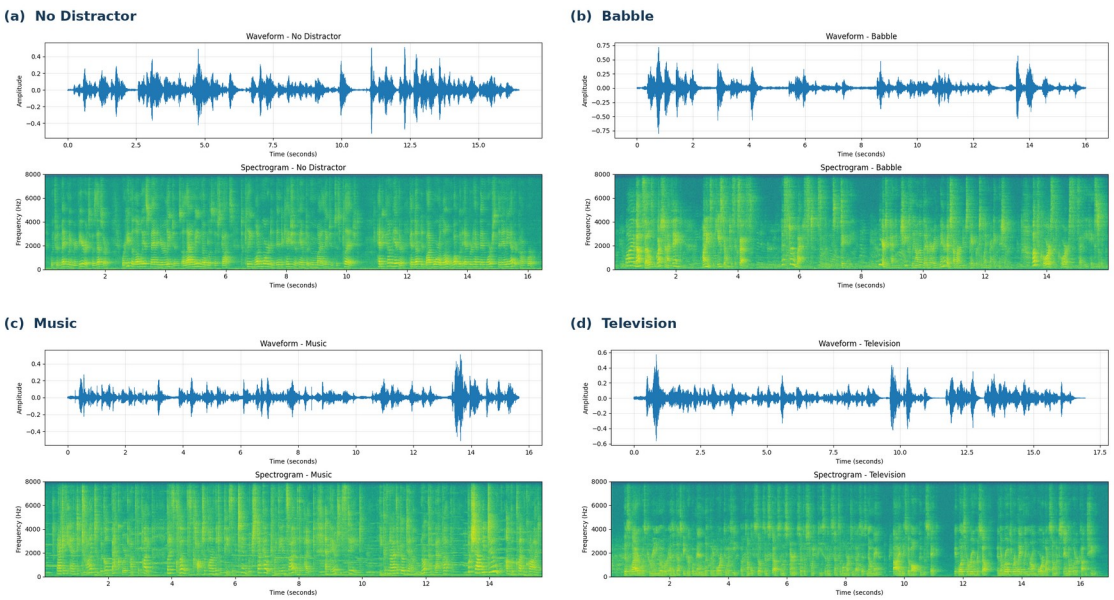


Figure 1. Waveform and spectrogram examples for the four background classes.

4. Background classification

4.1 Baseline features and evaluation

I extracted 13 MFCC means, 13 MFCC standard deviations, and the mean and standard deviation of RMS energy and zero-crossing rate (30 features per recording). I fitted an RBF-kernel SVM with C = 1 and gamma = scale in a pipeline with StandardScaler. I fitted the scaler on training data only and evaluated the classifier on the held-out speaker-disjoint test set.

Measure	Whole recording SVM
Accuracy	0.6942 (69.42%)
Macro precision	0.6932
Macro recall	0.6905
Macro F1	0.6901
Correct predictions	168 of 242

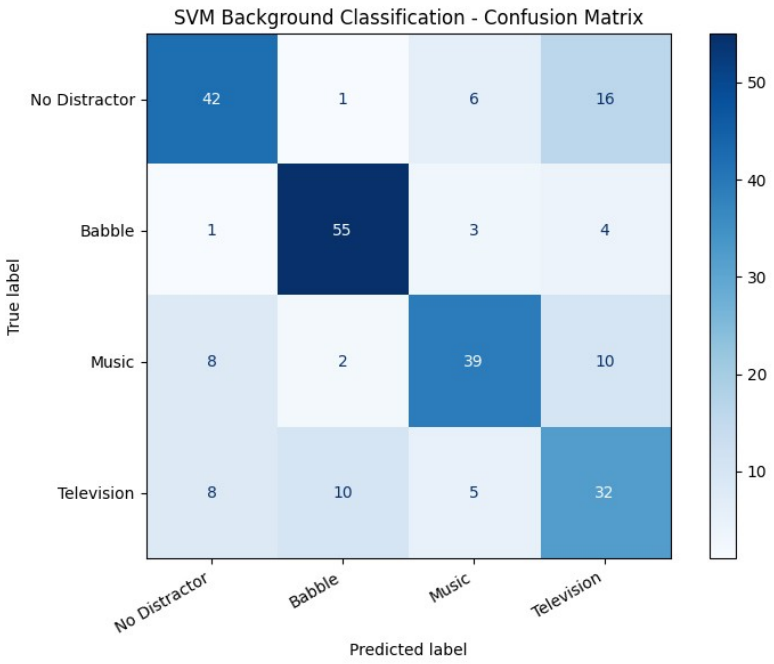


Figure 2. Baseline SVM confusion matrix on 242 held-out test recordings.

4.2 Low-energy feature selection

I tested whether background sounds were more distinguishable in less energetic regions of each recording. I selected the 25% of analysis frames with the lowest RMS values and calculated the same 30 acoustic statistics using those frames. I did not treat the selected frames as verified silent intervals. I retained the same SVM settings, labels, and speaker-disjoint split for a consistent comparison.

Metric	Whole recording	Low-energy frames
Test accuracy	69.42%	87.60%
Test macro precision	0.6932	0.8735
Test macro recall	0.6905	0.8733
Test macro F1	0.6901	0.8730
Correct predictions	168 / 242	212 / 242

I correctly classified 44 more test recordings with the low-energy feature set, corresponding to an accuracy improvement of 18.18 percentage points. In the low-energy confusion matrix, I obtained 55/65 correct predictions for none, 61/63 for babble, 54/59 for music, and 42/55 for television. I observed the lowest class recall for television in this evaluation.

4.3 Paired comparison and cross-validation

I compared the paired predictions for 242 test recordings. I found 160 recordings correctly classified by both models, 22 incorrectly classified by both, eight correctly classified only by the whole-recording model, and 52 correctly classified only by the low-energy model. My exploratory exact McNemar test gave a two-sided p-value of approximately 5.21×10^{-9}. I interpret this cautiously because recordings from the same speakers may be correlated and I examined the held-out results during exploratory feature comparison.

I also ran five-fold GroupKFold cross-validation on the 666 training recordings only, keeping speakers separate between each fold's training and validation partitions. I observed higher accuracy with low-energy features in all five folds.

Fold	Whole-recording accuracy	Low-energy accuracy
1	74.63%	91.04%
2	72.18%	93.23%
3	63.91%	81.20%
4	66.92%	82.71%
5	75.94%	90.98%
Mean	70.71%	87.83%

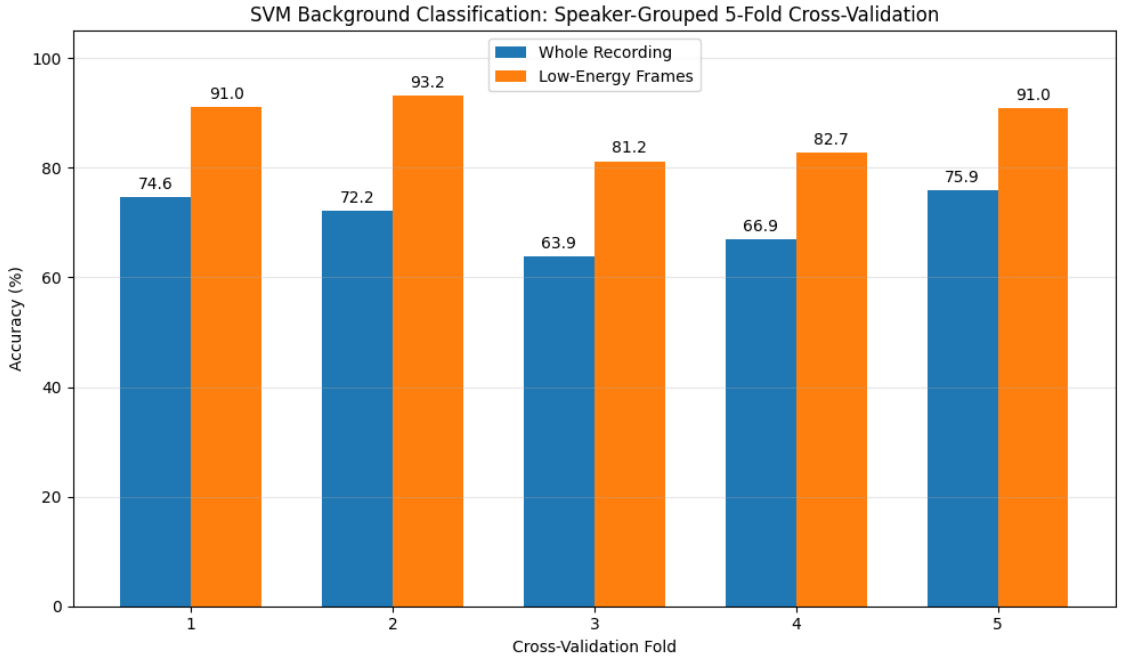


Figure 3. Five-fold speaker-grouped accuracy comparison using the training partition.

5. Voice activity detection

5.1 Methods

I implemented a classical energy detector using 25 ms frames and 10 ms hops. For each recording, I computed RMS energy in decibels, set a threshold at the 10th percentile plus 35% of the 10th-to-95th-percentile energy range, and applied 11-frame majority smoothing. I compared it with pretrained Silero VAD at 16 kHz. I used both methods to label speech versus non-speech at analysis-frame centers.

5.2 Four illustrative recordings

I first visualized one recording per background condition. I observed that Silero marked longer intervals as speech in all four examples. I measured frame agreement rates of 80.15% (none), 68.92% (babble), 66.58% (music), and 83.41% (television). I treated these recordings as illustrations rather than a controlled estimate of the background effect.

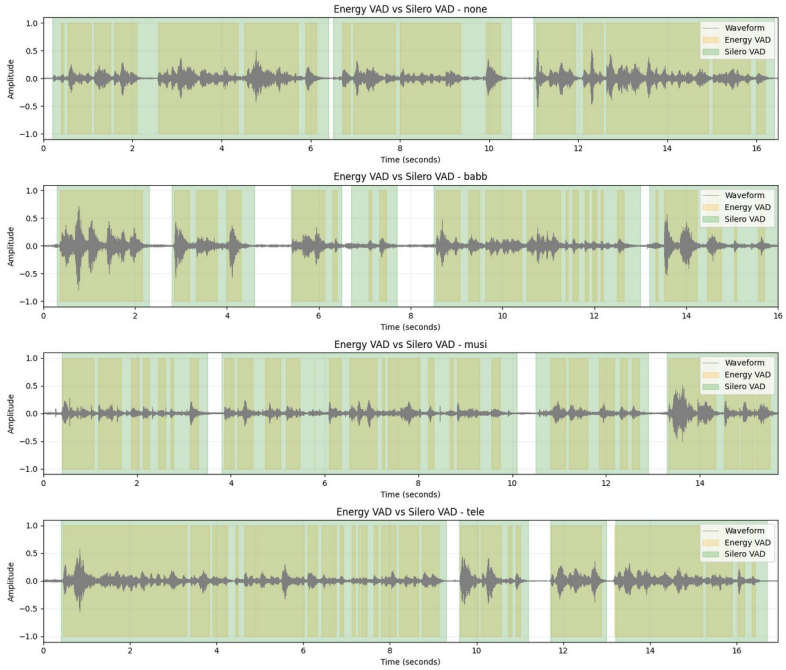


Figure 4. Comparison of detected speech regions for Energy VAD and Silero VAD in four example recordings.

5.3 Extended comparison: 40 recordings

I randomly selected 10 recordings from each background condition using seed 42. For each recording, I measured each method's detected-speech frame ratio, the frame agreement, and the direction of disagreement. I calculated the group means by assigning equal weight to each recording.

Background	Energy speech ratio	Silero speech ratio	Frame agreement	Agreement SD
No distractor	73.57%	83.48%	88.84%	5.09%
Babble	64.09%	89.16%	74.90%	9.59%
Music	63.75%	85.53%	76.44%	13.91%
Television	73.10%	85.24%	86.62%	6.62%
Mean across 40	68.63%	85.85%	81.70%	—

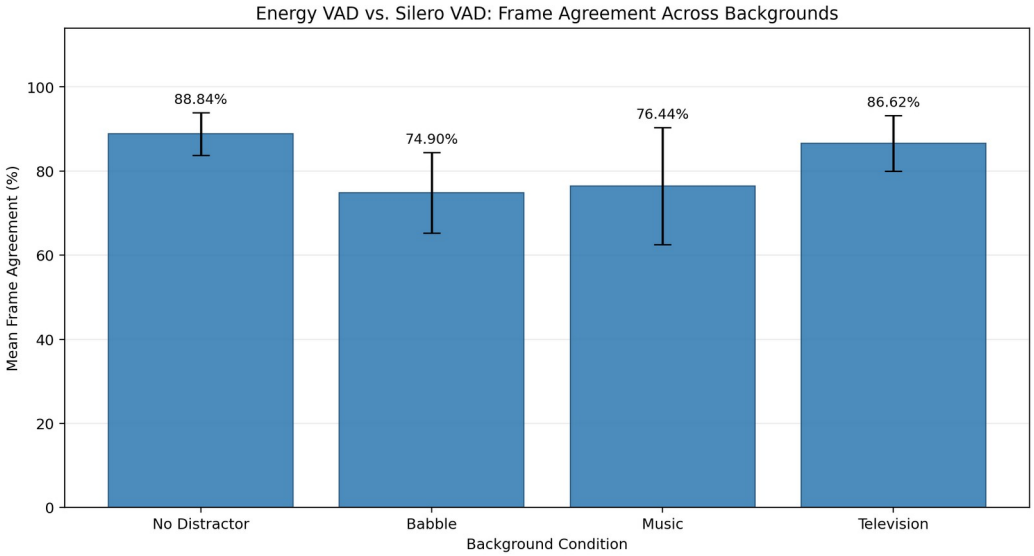


Figure 5. Mean frame agreement between VAD methods across background conditions; error bars show between-recording standard deviations (10 recordings per condition).

I found the highest mean frame agreement in the no-distractor condition and the lowest in babble. Across the 40 recordings, I observed that Silero classified a greater mean proportion of frames as speech than the energy detector. I interpret this as a difference in detector behavior, not as evidence that either method is more accurate.

6. Discussion and limitations

I observed the strongest classification improvement when I extracted features from the lowest-energy quarter of each recording. I considered a possible explanation: foreground speech contributed less strongly in some selected frames, making the background easier to recognize. However, I selected those frames by relative energy rather than by verified speech pauses. I therefore limit this interpretation to the VOiCES-subset recordings and the experimental conditions I evaluated.

I did not estimate the onset and offset of individual background events in the recording-level classification task. I identified probable speech intervals for VAD, but I had no independently annotated frame-level labels. For that reason, I did not report VAD accuracy, precision, recall, or F1 against ground truth. I compared 40 selected recordings without fully controlling for speaker, utterance, microphone, or room. I also treated the paired McNemar test as exploratory because recordings from the same speaker may be correlated.

I tested the complete notebook in a fresh Google Colab environment. I disconnected and deleted the previous runtime, connected to a new session, and used Run all. All 25 code cells completed without recorded execution errors. I saved the executed notebook and checked that its principal classification, cross-validation, and VAD results matched the values I report here.

7. Conclusion

I completed recording-level background classification and voice activity detection with VOiCES-subset. Using 30 acoustic features from whole recordings, I achieved 69.42% held-out accuracy with an RBF SVM. When I restricted the features to the lowest-energy quarter of the frames, I improved the accuracy to 87.60%. I observed the same direction of improvement in speaker-grouped cross-validation on training recordings (70.71% versus 87.83% mean accuracy). My comparison of Energy VAD and Silero VAD produced 81.70% mean frame agreement across 40 recordings. I conclude that feature selection substantially affected background classification in this experiment and that I must interpret VAD agreement cautiously in the absence of independently annotated speech intervals.

References

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